

Oluwatobi Melvyn Mayungbo

AI Engineer, LLM and Retrieval Systems

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SUMMARY

AI engineer who builds and deploys GenAI applications end to end. Built a retrieval assistant with hybrid search, guardrails and golden-set evaluation on GCP Cloud Run, an MCP server for AI assistants, and ML services on AWS ECS Fargate. Teaches machine learning to student cohorts.

SKILLS

AI and LLM Systems: Retrieval-Augmented Generation (RAG), LLM Evaluation and Guardrails, Hybrid Search (BM25, Dense Embeddings, RRF), Model Context Protocol (MCP), Hugging Face, Prompt Engineering, Citation Verification, ONNX Runtime

Machine Learning: PyTorch, Scikit-learn, LightGBM, XGBoost, Deep Learning, Sequence Models (GRU, TCN, Transformers), Model Calibration (Isotonic Regression, Temperature Scaling), SHAP, Feature Engineering

Engineering and MLOps: Python, FastAPI, Docker, GCP Cloud Run, AWS ECS Fargate, SQL, DuckDB, PostgreSQL, Redis, Railway, Prefect, dbt-core, MLflow, Evidently, CI/CD (GitHub Actions), Git, Pytest, Streamlit, React

Statistics and Causal Inference: Statistical Inference, Experiment Design, Power Analysis, CUPED, Causal Inference (S/T/X-Learners, Causal Forests), Uplift Modeling, Qini and AUUC, Time-Series Backtesting, Logistic Regression, WOE Scorecards

Data Analysis: Pandas, NumPy, SQL, Anomaly Detection

Payments and Risk: Transaction Fraud Detection, Credit Risk Decisioning, Payment Reconciliation, Double-Entry Ledgers and Idempotency, Regulatory Compliance (CBN/NDPA), Demand Forecasting, Web3 Fraud Risk Scoring

PROJECTS

Fintech Compliance RAG Assistant | [GitHub](#) | [Live App](#)

Tech: Python, Streamlit, BM25, ONNX Runtime, Reciprocal Rank Fusion, FastAPI, Docker, GCP Cloud Run

- Built and deployed a regulatory question-answering system over central bank (CBN) circulars and the NDPA, combining sparse (BM25) and dense (ONNX-served MiniLM) retrieval fused via reciprocal rank fusion, reaching 0.96 recall@5 and 0.842 MRR on a 50-question labelled golden set with zero retrieval misses in the top 10.
- Ran retrieval and generation evaluation as separate metrics to localise failure by component; traced a 0.40 refusal-correctness result to a lexical-overlap limitation in the offline stub judge and flagged re-validation against a live provider as the next step instead of overstating readiness.
- Designed generation to fail safely under uncertainty: a sentinel-token refusal mechanism enforced across the prompt, output guardrail and evaluation harness, plus post-generation citation verification that strips any marker the model cited but never retrieved.
- Deployed on GCP Cloud Run via Docker within a ~225MB peak memory footprint (2GB budget), with per-session and daily request caps in application code to bound API cost on a public endpoint.

web3-risk-mcp: Web3 Fraud Risk MCP Server | [GitHub](#)

Tech: Python, Model Context Protocol (MCP), Etherscan V2, GoPlus, DexScreener, Docker, GitHub Actions

- Built an MCP server that gives AI assistants (Claude Desktop, Cursor) six read-only tools to check a crypto wallet, token or smart contract for risk before anyone interacts with it, returning a 0 to 100 risk score with a reason for every point, across Ethereum, Base, Arbitrum One, Polygon PoS and BNB Chain.

Telco Customer Churn, End to End on AWS | [GitHub](#)

Tech: Python, XGBoost, MLflow, FastAPI, Gradio, Docker, GitHub Actions, AWS ECS Fargate, Application Load Balancer, CloudWatch

- Served it through a FastAPI /predict endpoint with a Gradio UI, containerised with Docker and deployed on AWS ECS Fargate behind an Application Load Balancer, with CloudWatch logs and GitHub Actions CI/CD.

Sequence-Based Transaction Fraud Detection System | [GitHub](#)

Tech: Python, PyTorch, LightGBM, ONNX Runtime, FastAPI, Docker, SHAP

- Deployed real-time scoring by exporting to ONNX Runtime (2.9× faster than eager PyTorch at batch size 1) and serving via FastAPI, reaching 5.1 ms p99 end-to-end latency against a 50 ms budget, with append-only per-decision audit logging and attention-based attribution of the prior transactions driving each flag.

EXPERIENCE

Machine Learning Instructor, SQI College of ICT March 2026 to Present

- Teach machine learning and deep learning to student cohorts, covering statistical foundations, classical ML algorithms, neural architectures, sequence models and production evaluation metrics.
- Instruct students on data analysis, feature engineering and model building with PyTorch, Scikit-Learn, Pandas and NumPy.

Internal Auditor, NISER May 2024 to July 2024

- Analysed financial and operational records to identify inconsistencies and irregularities, performing manual anomaly detection across complex transactional datasets.

EDUCATION

B.Sc. Statistics, University of Ibadan January 2021 to March 2025

Professional Diploma in Artificial Intelligence, SQI College of ICT July 2025 to Present

CERTIFICATIONS

- Google Data Analytics Professional Certificate
- Google Advanced Data Analytics Professional Certificate
- Claude Code In Action